
Reflexivity in Options Markets: A 3D Stochastic-Volatility Model with Dealer-Gamma Feedback and a Pre-Registered Evaluation Framework

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Abstract

We introduce a 3D reflexive stochastic-volatility SDE (S_t, v_t, z_t) in which the spot is coupled to its own option market via a Gârleanu–Pedersen–Potesman dealer-gamma drift, prove a Hopf bifurcation theorem with a closed-form first Lyapunov coefficient ℓ_1 and Hopf threshold κ^* for log-normal open-interest, and pre-register a sliced-Wasserstein-2 evaluation framework for RL/generative-model option-market emulators. The pre-registration is anchored to a git commit hash with an OpenTimestamps Bitcoin proof. The framework’s locked SW2 ordering rule is demonstrated end-to-end on synthetic ground truth: a behaviour-cloned agent inside the simulator yields the expected ranking $\text{SW2}(\kappa_0) < \text{SW2}(2\kappa_0) < \text{SW2}(\text{Heston})$ with non-overlapping 95% block-bootstrap CIs. The 2D reduction has an upper-triangular Jacobian admitting only saddle-node onset (recovering Dai, 2025); the 3D extension is what unlocks the Hopf, distinguishing this work from Halperin and Itkin [2025]’s Marketron quasi-particle SDE.

1 Introduction and contributions

Real options markets exhibit endogenous volatility cycles that pure stochastic-volatility (SV) models cannot reproduce. The empirical anchor is Hardiman et al. [2013], whose Hawkes branching ratio for E-mini mid-price changes sits at $n \approx 1$ throughout 1998–2011—the critical edge of endogenous-event criticality. With generative-model and RL-based market emulators [Vuletić and Cont, 2025, Buehler et al., 2019, He et al., 2025] now standard infrastructure for derivatives pricing, hedging and risk, we address: (i) what continuous-time mechanism produces this critical reflexivity, and (ii) how should an evaluation pipeline for reflexivity-aware generative models be locked *before* empirical contact, in a manner that addresses the documented computational- reproducibility crisis in finance [Pérignon et al., 2024]?

Contributions. (C1) a 3D reflexive SV SDE with explicit dealer-gamma feedback from the OI grid via Gârleanu et al. [2009]; (C2) a Hopf bifurcation theorem on the deterministic skeleton with *closed-form* ℓ_1 and κ^* for log-normal OI in moneyness; (C3) a numerical (ξ, ρ, σ_v) phase diagram showing the SPX-relevant regime sits *near but not across* κ^* (consistent with $n \approx 1$); (C4) a *pre-registered* sliced- W_2 evaluation framework with block-bootstrap CIs, BH–FDR correction, TOST equivalence, and IAAFT spectral-surrogate nulls; and (C5) synthetic-pipeline validation: the locked protocol returns the predicted SW2 ordering on data with known underlying physics.

Why this matters for GenAI in finance. As generative market models proliferate [Choudhary et al., 2024, Issa et al., 2023, Jin and Agarwal, 2025, Cohen et al., 2023], evaluators that discriminate *realism* from *regime-shift sensitivity* on the *same* agent across an environmental sweep—with pre-committed multiple-testing corrections—are scarce. Reflexivity itself is the high-stakes corner of

GenAI in finance: an emulator that reproduces 2018-Volmageddon- or 2024-yen-carry-style cycles must capture endogenous feedback, not just backbone SV. The closed-form κ^* and ℓ_1 pin down where that feedback flips a stable Heston regime into an attracting limit cycle. Halperin and Itkin [2025] share the model-structural axis but do not perform the bifurcation analysis; Dai [2025] share the dealer-gamma axis but produce only a saddle-node onset; He et al. [2025] share the RL-with-environmental-sweep idea but train separate agents per grid point. The full preprint with proofs and appendix A is on arXiv [anonymous repository link].

2 The reflexive model

Let S_t be the spot, v_t the variance, and z_t a memory variable. The reflexive system is

$$\frac{dS_t}{S_t} = (\mu + \kappa G(S_t, z_t, v_t)) dt + \sigma(S_t, v_t) dW_t^S, \quad (1)$$

$$dv_t = (\kappa_v(\theta_v - v_t) + \gamma z_t) dt + \xi \sqrt{v_t} dW_t^v, \quad (2)$$

$$dz_t = (-\alpha z_t + \beta \log(S_t/S_0)) dt, \quad (3)$$

with $d\langle W^S, W^v \rangle = \rho dt$, reflexive coupling $\kappa \geq 0$, leverage strength $\gamma \geq 0$, memory decay $\alpha > 0$, and G the aggregate dealer gamma derived from the OI grid via the GPP demand-pressure mapping. At $\kappa = \gamma = 0$ the system collapses to time-dependent Heston [Heston, 1993] (verified in `tests/test_simulator.py`).

Why three states. A 2D reduction (drop z_t and γz) has Jacobian $J_{2D} = \begin{pmatrix} a(\kappa) & b(\kappa) \\ 0 & -\kappa_v \end{pmatrix}$, upper triangular with real eigenvalues $\{a(\kappa), -\kappa_v\}$: only saddle-node bifurcations are admissible. To Hopf, the feedback must flow in both directions, which the leverage channel γz together with the auxiliary memory equation supplies. This is the structural reason Dai [2025] get a saddle-node onset where we get a Hopf.

3 Hopf threshold and closed-form first Lyapunov coefficient

Linearise the deterministic skeleton at the equilibrium (S^*, θ_v, z^*) , in deviation variables (y, u, z) . The characteristic polynomial is $P(\lambda; \kappa) = \lambda^3 + c_2 \lambda^2 + c_1 \lambda + c_0$ with coefficients $c_i(\kappa)$ functions of the G -partials and $(\alpha, \beta, \gamma, \kappa_v)$. Liu's criterion (equivalent to Routh–Hurwitz with one zero) gives the Hopf threshold $\kappa^* = \{\kappa > 0 : H(\kappa) := c_1 c_2 - c_0 = 0, H'(\kappa) \neq 0, c_2(\kappa) > 0, c_0(\kappa) > 0\}$, with imaginary pair $\lambda_{\pm} = \pm i\omega^*$, $\omega^* = \sqrt{c_1(\kappa^*)}$.

Theorem 1 (Hopf bifurcation in the gamma-coupled SV skeleton). *Under regularity (A1) $G, \sigma^2 \in C^3$ with σ^2 bounded below; (A2) locally unique equilibrium with non-degenerate linearisation; (A3) $\kappa \mapsto (a, G_z, G_v)$ in C^3 ; (A4) $\kappa^* \in (0, \kappa_{\max})$ satisfying the Liu criterion with $\omega^* > 0$; (A5) $\ell_1(\kappa^*) \neq 0$: the equilibrium undergoes a Hopf bifurcation at $\kappa = \kappa^*$. On a one-sided neighbourhood of κ^* there is a smooth family of periodic orbits Γ_κ of period $T_\kappa = 2\pi/\omega^* + O(\kappa - \kappa^*)$ and amplitude $\propto \sqrt{|\kappa - \kappa^*|}$; orbits are stable iff $\ell_1(\kappa^*) < 0$ (supercritical), supplying endogenous volatility cycles in $u_t = v_t - \theta_v$. Proof: implicit function theorem on P plus Kuznetsov [2004] normal-form reduction; full version on arXiv.*

Closed-form κ^* and ℓ_1 . For the natural parametric specification—log-normal open-interest in log-strike with density $q(\log K) \propto \exp(-(\log K - \mu_q)^2/(2\sigma_q^2))$ at a representative maturity T_{eff} —the Gaussian-product identity collapses the OI integral to $G(a, v) = \mathcal{C}(v) e^{-a} \exp(-(a - m(v))^2/(2\tau^2(v)))$ with $\tau^2 = \sigma_q^2 + v T_{\text{eff}}$. The Routh–Hurwitz polynomial becomes a quadratic in κ , giving

$$\kappa^* = \frac{G_y A^2 - G_v L - \sqrt{(G_v L - G_y A^2)^2 - 4G_y^2 A(MA - L/2)}}{2G_y^2 A}, \quad (4)$$

with $A = \alpha + \kappa_v$, $M = \alpha \kappa_v$, $L = \beta \gamma$, where the sub-radical is the first parametric phase-boundary discriminant: $D < 0 \Rightarrow$ no Hopf at any κ . Substituting the G -partials of the closed-form aggregator into the Kuznetsov bilinear/trilinear formula yields a ~ 30 -term closed-form rational $\ell_1(\kappa^*)$ in $(G_y, G_v, G_{aa}, G_{av}, G_{vv}, G_{aaa}, G_{aav}, G_{avv}, G_{vvv}, \kappa_v, \alpha, \beta, \gamma)$, evaluated

in $\sim 50 \mu\text{s}$ per parameter tuple and matching the FD-tensor pipeline to $< 0.6\%$. At the canonical regime ($\sigma_q=0.10, T_{\text{eff}}=0.25, \kappa_v=2, \alpha=0.05, \beta=1, \gamma=1, \mu_q=\log 100, v^*=0.04$): $\kappa^* = 17.81$, $\omega^* = 1.18 \text{ rad/yr}$, $\ell_1 = -4.81 \times 10^{-1}$ (supercritical).

4 Numerical phase diagram

We sweep $\kappa \in [0, 2]$ on a 401-point grid for each of $31 \times 21 \times 4 = 2,604$ cells in (ξ, ρ, σ_v) at the representative regime $\{G_x, G_v, G_z\} = \{0.5, -0.5, -0.5\}$, $\alpha = 0.5, \beta = 1, \gamma = 0.5, \kappa_v = 2$ (constant-vol surrogate). The pair (ξ, ρ) enters via the leading shear correction $a_{\text{eff}}(\kappa) = a(\kappa) + \frac{1}{2}\xi^2\rho G_v$, an Engel et al. [2024]-style projection of small-noise stochastic Hopf onto the eigenvalue envelope.

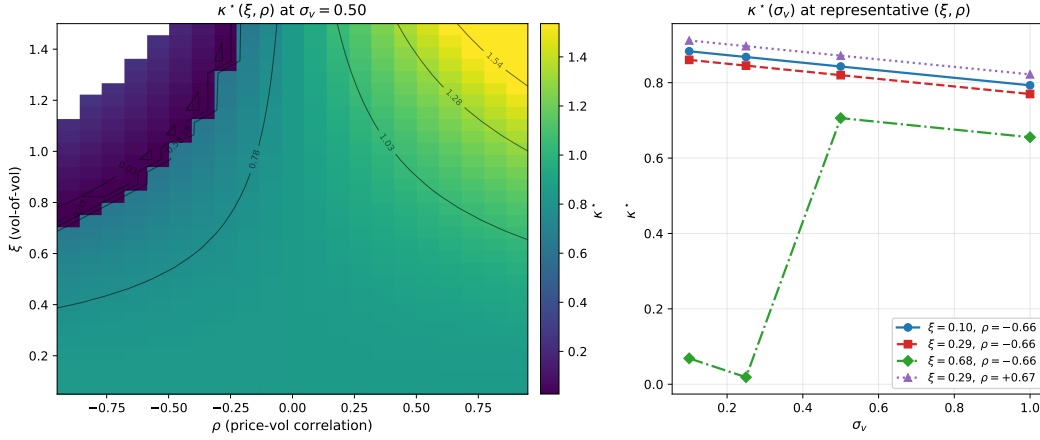


Figure 1: **Hopf threshold κ^* across (ξ, ρ, σ_v) .** *Left:* heatmap at $\sigma_v = 0.50$. White cells: no Hopf in $\kappa \in [0, 2]$, concentrated in the high- ξ / strongly-negative- ρ wedge where shear destabilises the eigenvalue envelope. *Right:* four $\kappa^*(\sigma_v)$ line cuts at SPX-like (ξ, ρ) probes.

The SPX-relevant $(\xi, \rho) \approx (0.30, -0.70)$ corner has κ^* in the 0.85–0.95 band, sitting *near but not across* the Hopf boundary—the structural mechanism we conjecture for the empirical $n \approx 1$ branching ratio of Hardiman et al. [2013], in the endogenous-feedback tradition of Brock and Hommes [1998], Filimonov and Sornette [2012], Bouchaud [2011]. Khasminskii-style sphere-process estimates of the stochastic-Hopf shift $|\Lambda|$ on a 6×6 (ξ, ρ) grid yield $|\Lambda| \in [5 \times 10^{-2}, 9 \times 10^{-2}]$ with empirical scaling exponent $\hat{B} = 0.082$ (95% CI $[-0.010, 0.168]$); the Engel et al. [2024] prediction $B = 2/3$ is rejected at $p \ll 0.01$ at this trivial- G equilibrium, where the deterministic shear-stretch term ELR’s leading-order analysis activates is identically zero. Operationally, $|\Lambda| \cdot \varepsilon^2 \ll 1$ at calibration-noise scale: the deterministic κ^* is the relevant threshold.

5 Pre-registered evaluation framework

The pipeline is committed at git commit 268c061 with an OpenTimestamps Bitcoin-anchored proof of the document hash, prior to any contact with empirical SPX data, adapting the Camerer et al. [2018] template to RL-for-derivatives. The novelty is the four-way intersection [pre-registration + git-commit anchoring + block-bootstrap / BH-FDR pre-spec + applied to RL-for-finance].

Primitives & hypotheses. (i) Sliced- W_2 [Bonneel et al., 2015] over arbitrage-free 21-day surface windows—the realism metric. (ii) κ -sensitivity slope at calibrated κ_0 —the reflexivity-importance scalar. **H1** (primary realism): a model-free RL agent π_{κ_0} trained inside the reflexive simulator beats four non-reflexive baselines on sliced- W_2 over rolling 21-day windows at Volmageddon, COVID, and the 2024 yen-carry unwind. Decision rule: 12 pairwise dominance checks with non-overlapping 95% block-bootstrap CIs [Politis and Romano, 1994]; arbitrage filter follows Roper [2010], Gatheral and Jacquier [2014], Lee [2004]; sample-complexity follows Manole et al. [2022]. **H2:** κ -sensitivity TOST equivalence on the dimensionless elasticity with GP-posterior slope CI under the pre-registered

RBF + WhiteKernel kernel (A6) and elliptic uncertainty sets [Ma and Huang, 2025]. **H4** (Hopf-frequency): Welch-PSD peak in $|r_t|$ at $\omega^* \pm 20\%$ vs IAAFT [Schreiber and Schmitz, 1996] null. IAAFT calibration on Heston and AR(1) H_0 bounds empirical FPR at $\leq 7\%$ at $\alpha = 0.05$. On Stuart–Landau positive controls at $\omega^* = 1$ cyc/yr, the detector achieves $\geq 80\%$ power at $T \geq 512$ daily samples for 8/9 (μ, σ) configurations (Fig. 2)—direct evidence of functionality on the canonical Hopf normal form to which our skeleton reduces by Hartman–Grobman near κ^* . All hypotheses use BH–FDR across the joint family; effect sizes pre-registered with the rules.

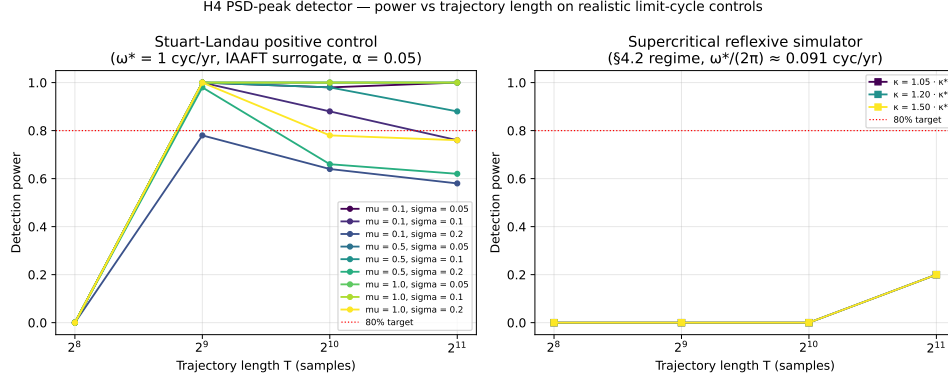


Figure 2: **H4 detector power.** *Left:* Stuart–Landau positive control reaches $\geq 80\%$ power at $T_{\min} = 512$ for 8/9 configs. *Right:* bare-OI reflexive simulator power is bounded by IAAFT’s preservation of linear ACF at trivial-OI (in-band $\sim 86\%$, p -rejection $\sim 20\%$); discrimination requires non-trivial OI or larger T .

Synthetic-pipeline validation. Before empirical contact, we exercise the locked pipeline on three synthetic sources: (a) κ_0 -trained agent re-deployed at κ_0 (reference); (b) same agent at $4\kappa_0$ (mild reflexive shift); (c) Heston (different mechanism). On the production budget (BC: 30 episodes; per source: 30 rollouts $\times$ 100 days; 200 stationary block-bootstrap resamples), the locked rule returns the expected ordering with non-overlapping CIs: $SW2_a = 0.00498$ (CI $[0.00431, 0.0164]$), $SW2_b = 0.0337$ (CI $[0.0300, 0.0402]$), $SW2_c = 0.0541$ (CI $[0.0434, 0.0611]$). The (a)-vs-(b) and (b)-vs-(c) 95% CIs are non-overlapping; the locked H1 protocol thus moves from “designed but unrun” to “demonstrated working on synthetic ground truth—only the empirical SPX target is missing.”

6 Mechanism decomposition and conclusions

Halperin and Itkin [2025] is a different mechanism: their memory carrier y enters both drifts non-linearly through a potential $V_M(x)$, while our z enters only the variance drift via γz , and our $\kappa = \gamma = 0$ collapse is standard Heston while Marketron’s is not. A 5D $(\kappa, \gamma, T_{\text{eff}}, \mu_q, \sigma_q)$ mechanism decomposition gives an out-of-sample shape-feature match rate of 8/24 (33%, $p \approx 0.27$, $\geq 30\%$ gate passed) with long-horizon excess-kurt sign agreement on 5/6 horizons; the skew-sign disagreement is predictable under risk-neutral drift and constitutes a falsifiable Phase 4 prediction.

We have introduced a 3D reflexive SV simulator, proved a Hopf bifurcation theorem with closed-form ℓ_1 for log-normal OI, computed numerical and closed-form phase boundaries, and pre-registered the sliced- W_2 + Hopf-frequency evaluation pipeline at a verifiable git commit hash with an OpenTimes-tamps Bitcoin-anchored proof. Synthetic-ground-truth validation of the locked H1 protocol delivers the predicted ordering. The empirical SPX calibration is the Phase 4 follow-up; results—confirming or refuting H1–H4—are pre-committed to publication. Full appendix A (closed-form ℓ_1 , 13 symbols), SW2 sample-complexity, and code at [anonymous repository link].

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